ARFAT NAIK

Data Science & Machine Learning Aspirant

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SUMMARY

Aspiring Data Scientist with hands-on experience in end-to-end machine learning projects, model deployment, and applied NLP. Skilled in Python, scikit-learn, Pandas, and Flask, with demonstrated results on real-world datasets. Seeking a Data Science / Machine Learning internship.

TECHNICAL SKILLS

Programming & Data: Python (NumPy, Pandas) , SQL
ML: Supervised Learning (Regression, Classification), Feature Engineering, Model Training & Model Evaluation, Cross-Validation, Hyperparameter Tuning, Performance Metrics (ROC-AUC, Precision, Recall, F1-Score)
Data Science: Exploratory Data Analysis (EDA), Data Visualization (Matplotlib, Seaborn), Statistical Analysis, Data Cleaning, Outlier Detection, Missing Value Imputation
Tools & Frameworks: Git, GitHub, Flask, Joblib, VS Code, Jupyter Notebook
NLP / Recommendation Systems: TF-IDF Vectorization, Cosine Similarity, Content-Based Recommendation Systems
Databases & Caching: PostgreSQL (Supabase), Redis (In-memory store)

PROJECTS

AtmosIQ

• Built an end-to-end air quality forecasting system using a **Random Forest regression model** to predict AQI 3 hours ahead with **$R^2 \approx 0.85$**, leveraging lag features, rolling statistics, and meteorological data.
• Designed an automated data pipeline using **GitHub Actions, Redis, and Supabase** to ingest real-time pollution data hourly, perform feature engineering and inference, and persist predictions for historical analysis.
• Developed and deployed a **production-grade Flask dashboard** with interactive visualizations (Chart.js) to monitor PM2.5, PM10, current AQI, and short-term forecasts, hosted on **Railway**.
Repository: https://github.com/Arfatnaik0/AtmosIQ
Live link: https://atmosiq.up.railway.app/

Credit Risk Prediction System

• Developed a credit risk prediction ML system using **Logistic Regression** and **cost-sensitive thresholding** to optimize loan approval decisions.
• Engineered features, prevented data leakage, **imputed missing values**, and **managed class imbalance** for robust model evaluation.
• Deployed model via **Flask** for real-time default probability estimation and decision output.
Repository: https://github.com/Arfatnaik0/credit_risk_prediction_system
Live link: https://credit-risk-prediction-system.vercel.app/

Movie Recommendation System.

• Developed a content-based recommender for **5,000+ TMDB** movies using **TF-IDF** and **cosine similarity**.
• Engineered an NLP feature pipeline that combines genres, keywords, cast, director, and overview.
• Built Flask web app with **real-time recommendation API** and input validation.
• Optimized similarity computation using sparse matrices, reducing runtime by **60%**
Repository: https://github.com/Arfatnaik0/movie_recommendation
Live Link: https://movie-recommendation-ai22.onrender.com/

EDUCATION

Bachelor's Degree in Electronics and Computer Engineering
Rizvi College of Engineering (2023–2027)
Semester 6 | CGPA: 7.27

Certifications

Programming, Data Structures and Algorithms using Python – NPTEL
Certificate- https://drive.google.com/file/d/1-6FNNG1wNs6x1pBJ9LSpyjDSFKkMzJts/view?usp=sharing